Part 1 // Artificial Intelligence

Artificial Intelligence (AI): Develop machines with the ability that are usually done by us human with our natural intelligence. (e.g. Siri or Alexa)

Machine Learning (ML): Enables the computer to learn on their own by analysing a vast amount of data without hand-coded instructions. (e.g. Netflix's recommendation system)

Deep Learning (DL): A subfield of machine learning. Mimics the activities of neurons in our brains with deep neural networks. (e.g. GenAI: ChatGPT, Claude, Midjourney, Stable Diffusion, etc)

Relationship:

- **AI** is the overarching concept that encompasses both ML and DL.
- **ML** is a specific approach within AI that focuses on learning from data.
- **DL** is a more advanced form of ML that uses neural networks with many layers to analyze data and learn patterns.

AI applications:

1.Disease detection & Cancer diagnosis: assisting doctor to diagnose

➔Reduce examination workload

➔Increase the number of misdiagnosed patients

2.AlphaFold: Knowing the shape of a disease-related protein can aid in development drugs and treatments

➔ Lower the cost

➔Allows researchers to target diseases that are often neglected due to high costs.

3.AI application - Smart hospitals in HK

The Medical Centre of Chinese University of Hong Kong

➢ cameras to analyse patient's physical condition

➢ Data collected are helpful for rehabilitation training

Tin Shui Wai Hospital (2017)

➢ chest X-rays, CT scans, endoscopy, and hip fracture detection

➢ AI is used in brain CT scans, endoscopy, and hip fracture detection

4.Self-driving cars

L0: No Automation, driver perform all

L1: Driver Assistance, assist either in steering, acceleration or braking

L2: Partial Automation, assist both in steering and acceleration or braking

L3: Conditional Automation, vehicle perform all driving, but with human intervention

L4: High Automation, vehicle perform all driving tasks without human intervention under specific conditions

L5: Full Automation, vehicle perform all

(e.g. Baidu's Apollo Go, August 2022, lv4)

Subfields of AI

1.Computer Vision: enables computers to derive meaningful information.

➢ Image segmentation: label specific regions with a corresponding class of what is being represented.

➢ Sam2 (Meta AI): precise selection of any object in any video or image.

2.NLP: ability to understand text and spoken words in much the same way human beings can. (E.g. Translation, Chatbot)

➢ Speech recognition: Speech to text (E.g. Whisper)

➢ Speech synthesis: Text-to-Speech (E.g. Elevenlab)

- Face Detection is about finding faces.
- Face Analysis extracts information about facial expressions and demographics without identifying the person. (E.g. age, gender, emotion detection)
- Face Recognition is about identifying or verifying the individual based on facial features.

What is most important today: creativity, problem solving, soft skills and self-learning

T shape skill: Deep expertise in one area, along with broad understanding of other areas

M shape skill: Multiple areas of expertise, along with broad understanding of other areas

Big Data and Data Analytics

4V of Big Data

- Volume: A huge amount of data
- Velocity: High speed and continuous flow of data
- Variety: Different types of structured, semi-structured and unstructured data coming from heterogenous sources
- Veracity: Data may be inconsistent, incomplete and messy

Type of data:

1. Structured data: a data model or schema and is often stored in tabular form. (E.g. table)
2. Semi-structured data: Non-tabular structure but conform to some level of structure. (E.g. email, html)
3. Unstructured data: not conform to a data model or data schema. (E.g. video, image and audio)

4 types of data analytics

1. Descriptive Analytics: What happened?
2. Diagnostic Analytics: Why it happened? (Cause of a phenomenon that occurred in the past.)
3. Predictive Analytics: What will happen? (Generate future predictions based upon past events.)
4. Prescriptive Analytics: What should I do? (Make smart decisions)

******Machine Learning***

Earlier Ai: Elisa (rule-based Ai), Expert systems

What is machine learning: learning from data, human do not need to define the rules

Types of machine learning:

Supervising learning: using a label data set, given an image ai can learn the relationship between image and label

Classification models: Use attributes ($x1, x2, \ldots.$) to predict a categorical variable (y)

➢ Spam Detection: Predict whether an email is spam/ham based on words and the attachment.

➢ Sentiment analysis: Predict whether a movie review is positive or negative, based on the words in the review.

➢ Image recognition: Take as input the pixels in the image, and they output a prediction of what the image depicts.

Difference between binary and multiclass classification

Aspect	Binary	Multiclass
Number of Classes	2	More than 2
Examples	Spam/Not Spam, Cancer/Non-Cancer	Animal type (e.g., dog, cat, bird)
Complexity	Relatively simpler	More complex due to multiple categories
Training Data	Requires less data	Requires more data

➢ **Underfitting:**
Definition: Occurs when a model is too simple and cannot capture the underlying pattern in the data.
Example: Like not having studied enough for an exam—it's unprepared to perform well.

➢ **Overfitting:**
Definition: Happens when a model performs well on training data but fails to generalize to new, unseen data.
Example: Like cramming for an exam and only knowing the specific questions you studied—fine for the test, but not for new challenges.

➢ Ai writing detectors Major concern about educator: false positive

➢ **Measure the correctness or accuracy of the model:**

➢ **Confusion matrix:** is a table that categorizes predictions according to whether they match the actual value

Regression models: Use attributes ($x1, x2, \ldots.$) to predict a numerical variable (y)

➢ predict a numeric variable (target variable is numeric) the output is class/ category

$$Accuracy = \frac{No.\ of\ Correct\ Predictions}{Total\ Predictions}$$

True positive (TP): Actual AI-generated text that was correctly predicted as AI-generated text.

True negative (TN): Actual human text that was correctly predicted as human text.

False positive (FP): Actual human text that was incorrectly predicted as AI-generated text.

False negative (FN): Actual AI-generated text that was incorrectly predicted as human text.

Unsupervised learning: Extract information from a dataset that has no labels, or targets to predict.

Reinforcement learning: no labelled data is given.

E.g. training robot to do some tasks, car parking and how to walk

AlphaGO(chess ai)

Deep learning

	Traditional ML	DL
Feature Extraction	Requires manual feature extraction and selection.	Automatically extracts features through layers of neural networks.
Training Data	Performs adequately with smaller datasets.	Requires large amounts of data to perform effectively.
Computation power (training)	Lower	Higher
Interpretability	Models are usually more interpretable and easier to understand.	Models are often considered "black boxes" and harder to interpret.

Traditional machine learning algorithms do not gain in accuracy beyond a certain amount of training data.

Deep learning is the first class of algorithms that is scalable

➔ Results get better with more data + bigger models + more computation

Deep learning requires more data and more computational power resources

Neural network to do the prediction

MNIST data set: handwriting recognition (1990s)

Neural network architecture: Fully connected feed-forward neural network.

Convolutional Neural Network (CNN): a neural network which utilizes a special type of layer (convolutional layers) to learn from image and image-like data.

Transformer: A widely used architecture in natural language processing tasks. (Foundation for generative AI models like GPT)

The meaning of the deep: the neural has many layers

Many layers➔learn more complicated patterns step by step

Keypoint detection: Involves identifying certain keypoints or landmarks on an object and tracking that object.

Pose Estimation: Detect the position and orientation of human body parts in images or videos

Prosthetics: electrical signals produced by muscle activity

Part 2 // AI Robotics

Definition of Robot: Robotics derives from "robot" (Slavic "robota," meaning servant). Introduced by George Devol as artificial people (e.g., humanoid robots).

Key Characteristics:

- Sensing: Ability to perceive environment (light, touch, sound, etc.).
- Movement: Navigate surroundings for action.
- Energy: Self-charging capabilities.
- Intelligence: Basic knowledge to perform tasks.
- Applications: Widely used due to cost-saving, efficiency, and error reduction

AI Milestones (2014-2023)

GANs Invention (2014), Sophia the Robot Activation (2016), WaveNet AI Audio Generator Launch (2016), AlphaGo's Go Champion Win (2016), Birth of Deepfakes (2017), Development of Transformers (2017), OpenAI GPT-1 Launch (2018), AlphaFold's Protein Folding Contest Win (2020), Generative AI Mainstreaming (2022) , **ChatGPT-4 Release (2023)

Classification of Robots

The Japanese Industrial Robot Association (JIRA):

1. Manual handling device: Controlled by an operator with multiple degrees of freedom.
2. Fixed sequence robot: Executes tasks in a set, unchangeable sequence.
3. Variable sequence robot: Similar to Class 2, but the sequence can be easily modified.
4. Playback robot: Records a human's manual task and repeats it based on the recorded data.
5. Numerical control robot: Programmed by an operator to perform tasks.
6. Intelligent robot: Can understand its environment and complete tasks despite changing conditions.

Robotics Institute of America (RIA):

Type A: Handling devices with manual control to telerobotics.

Type B: Automatic handling devices with predetermined cycles.

Type C: Programmable, servo-controlled robots with continuous or point-to-point trajectories.

Type D: Similar to Type C but with the ability to acquire information from its environment.

Applications of robots:

- Humanoid Robots: Emergency response and caregiving.
- Industrial Robots: Assembly and hazard management.
- Medical Robotics: Drug development and surgeries.
- Service Robots: Assistance in hotels, shopping, and eldercare.
- Defense Robotics: Surveillance and combat

Advantages of Robots:

- Continuous operation.
- Error minimization.
- Work in hazardous environments (e.g., space, radiation zones).
- Increased productivity

Limitations of Robots:

- Limited creativity, decision-making.
- High initial and maintenance costs.
- Challenges in real-time response and adaptability

Elon Musk views on Optimus Humanoid Robot:

- Computing Power & Data: Efficient AI use and real-time data collection are crucial.
- Reality's Scale: Optimus learns from vast real-world data.
- Production Demand: High demand, potentially billions produced annually.
- Technical Design: Complex engineering needed for hand design, mimicking human muscles.

Industrial Manufacturing Development Stages:

First Industrial Revolution (1784):

- Steam engines as prime movers.
- Visual: Steam engine.

Second Industrial Revolution (1913):

- Mass production increases productivity.
- Visual: Assembly line with cars.

Third Industrial Revolution (1969):

- Programmable logic controllers (PLCs) for computational logic.
- Visual: Computer and robotic arm.

Fourth Industrial Revolution (2014):

- Use of information/intelligent technology in manufacturing.
- Visual: Robotic arm, computer, tablet, and smartphone.

Intelligent Manufacturing Cycle:

- Data: Collect data.
- Analytics: Analyze data to make predictions.
- Action: Use predictions to enhance the process.
- Control: Implement control actions.
- Product: Improve product quality.

Typical Robotic Applications in the Industry:

Arc Welding, Palletizing**,** Die Casting**,** Stamping, Machining, Sanding/Polishing & Deburring, Painting, Packaging, Injection Molding, Testing & Inspection, Assembly, Laser Marking, Welding & Cutting

AI Ethics

Three key factors enable AI success:

- Algorithms and Architecture: Innovative combinations of algorithms and architecture solve specific problems.
- Computing Power: Increased computing power has made previously impractical algorithms, like Deep Learning, viable.
- Big Data: Access to large amounts of data in the cloud supports Deep Neural Networks.

Computers excel at: Template matching, Numerical computation, Dealing with large datasets.

Humans excel at: Dealing with poor input quality, Flexibility, Common sense, Creativity.

Normal Accidents' and AI Safety:

- Inevitable Accidents: In highly complex systems with tightly coupled interacting elements, accidents are inevitable.
- Role of Organizations: Organizations play a critical role in creating and mitigating risks.
- AI Systems Risk: AI systems are prone to "normal accidents" due to their complexity.
- Regulation and Ethics: Technical safety in AI is closely tied to human regulation, value judgments, and ethics.

Benefits of AI	Risks of AI
Better recommendations	Filter bubbles
Safer cars	Fake news
Better medical diagnosis	Unfair matching

Autonomous driving – German ethics commission:

- Legal Scholars: To ensure AVs comply with existing laws and regulations.
- Ethics Experts: To navigate moral dilemmas and ensure AVs align with human values.
- Engineers: To provide technical insights into feasibility and implementation.
- Consumer Protection Groups: To safeguard user rights and interests.
- Religious Leaders: To consider broader moral and spiritual perspectives.

Overall Distribution of Ethics Theories (Top 3):

Utilitarianism: 30%, Rights: 12%, Kantian/Deontology: 10%

The Twelve Ethical Principles for AI:

1. Accountability: Explain AI decisions clearly to humans.
2. Human Oversight: Allow users to control AI decisions.
3. Transparency: Disclose AI use and data practices clearly.
4. Data Privacy: Protect individuals' privacy in AI development and use.
5. Fairness: Treat individuals equally, without bias.
6. Beneficial AI: Ensure AI benefits humans and prevents harm.
7. Reliability & Security: Develop reliable AI and protect from cyberattacks.
8. Diversity & Inclusion: Respect all stakeholders' interests.
9. Lawfulness: Comply with laws and regulations.
10. Safety: Ensure AI does not compromise human safety.
11. Cooperation: Foster open cooperation in the AI ecosystem.
12. Sustainability: Manage AI's societal and environmental impacts.

ISO/IEC 4200: Developed by ISO and IEC for AI

- AI Management System (AIMS): Certifiable framework for AI development and deployment within an AI assurance ecosystem.
- Goal: Ensure responsible AI use, benefiting organizations and society while reassuring stakeholders.

benefits of AIMS:

- Safe AI Implementation: Ensures responsibility and accountability.
- Security & Quality: Maintains safety, fairness, transparency, and quality.
- Strategic Decision: AI as a strategic move with clear goals.
- Strong Governance: Demonstrates robust AI governance.
- Governance & Innovation: Balances control and innovation.
- Responsible Use: Ensures AI's responsibility, continuous learning.
- Safeguards: All necessary protections in place.
- Process Implementation

AI strategies and policy focuses:

USA: Prioritizes economic growth, technological advancement, and national security.

Japan: Aims to build "Society 5.0."

EU: Focuses on ethical risks such as security, privacy, and human dignity.

China: Emphasizes industrializing AI applications for its "Manufacturing Power" strategy.

AI development Plan:

2020: AI technology at an advanced global level.

2025: Major breakthroughs in AI theory, tech, and applications, reaching world-leading status.

2030: AI achieves world-leading status in theory, tech, and applications, becoming a major AI innovation center globally.

Knowledge Graph:

The Knowledge Graph is a knowledge base used by Google to enhance its search engine's search results with semantic-search information gathered from a wide variety of sources.

Seach Technology:

Blind search: Depth-first search & Width first search

Heuristic search: A * / A Star Algorithm

iterated Local Search (ILS): Hill-Climbing

Adversarial Search: Alpha-Beta Pruning & Monte Carlo Tree Search

Expert System:

An intelligent computer program that uses knowledge and reasoning to solve complex problems that only experts can solve. (Edward Feigenbaum, 1982)

Generative AI Vs NLP Vs LLM:

Generative AI = Creativity, NLP = Understanding,
LLM = Advanced NLP

Definition of Big Data:

Data: Information in a formalized manner for communication, interpretation, or processing.

Dataset: Identifiable collection of data available for access or download in various formats.

Big Data: Extensive datasets with characteristics like Volume, Variety, Velocity, and/or Variability requiring scalable technology for efficient storage, manipulation, management, and analysis.

Five key characteristics of cloud computing:

Elastic Capacity: Scale resources up or down quickly based on demand. || Faster Time to Market: No infrastructure barriers; easy global deployment. || No Initial Investment: No upfront costs or long-term commitments. || Pay-as-you-go: Pay for what you use (servers, storage, data transfer). || Focus on Business: Less time on repetitive tasks, leverage AWS scale.

Artificial Intelligence of Things (AIoT):

Alot(accurate and clean data, stable sensor module and IoT platform, correct AI computing module to import data, cloud or edge intelligent computing system, choose an IoT PaaS platform (Platform as a Service) that can quickly develop AIoT application management functions